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Research paper

Hybrid trajectory planning and tracking for automatic berthing: A grid-search and optimal control integration approach

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ABSTRACT

Automatic berthing is considered one of the most challenging problems for underactuated unmanned surface vehicles (USVs) due to the influence factors such as rudder effect, environmental disturbances, and control uncertainties. This paper focuses on trajectory planning and tracking to address this problem. The hybrid approach initially employs a reverse search strategy to plan the global berthing trajectory, with the terminal part of this trajectory subsequently being reconstructed and optimized by taking into account environmental loads and USV dynamic model. A nonlinear MPC controller is designed for real-time tracking of the berthing trajectory. The rudder effect constraints related to USV speed is supplemented to the dynamic model to alleviate the problem of poor maneuverability at low speed. The whole trajectory planning and tracking scheme of automatic berthing is simulated at Dalian Port considering environmental loads and uncertainties in control. The results show that the scheme has quite fast solving speed of trajectory planning and robust control of trajectory tracking.

1. Introduction

Due to the growing demand in both military and civilian fields, coupled with the robust capability of self-planning and self-navigation in complex environments, unmanned surface vehicles (USVs) have garnered extensive research in the field of ocean engineering. The autonomous navigation for ships in open waters has gradually refined and has been preliminarily applied in engineering both domestically and internationally. However, the automatic berthing, which is the "last mile" for USVs to achieve complete autonomous navigation, is still immature, and this is also a scientific challenge recognized internationally.

Automatic berthing technology demands that USVs can autonomously make decisions on navigation paths, headings, and speeds without human intervention, and complete operations such as turning, steering, lateral movement, and stopping. For underactuated USVs, these requirements pose a significant challenge. Underactuated USVs are typically equipped with only one propeller and one rudder, lacking lateral thrusters, thereby constraining the maneuvering performance (Liu et al., 2022). The rudder effectiveness of the USV will gradually diminish or even be lost as the speed decreases during berthing (Liu et al., 2021). Moreover, the low-speed motion control of the USV is uncertain and affected by environmental loads such as wind, wave, and current. These factors collectively set higher demands for automatic berthing, especially in complex and narrow ports.

1.1. Literature review

Numerous studies have conducted extensive research on the automatic berthing maneuvers (Shimizu et al., 2022; Suyama et al., 2022; Peng et al., 2023; Higo et al., 2023; Liu et al., 2024; Sun et al., 2024). Among them, trajectory planning and tracking is one of the most commonly used control methods to realize automatic berthing. Here, a trajectory refers to the sequence of spatiotemporal states (i.e. time-varying waypoints) conforming to the constraints of USV dynamics (Claussmann et al., 2019). For an underactuated system, an important task of trajectory planning is to find a reference trajectory that can be precisely tracked by the USV itself. In addition to this, the berthing trajectory also requires that the speed of the last waypoint is zero and that the desired heading for berthing is achieved.

The berthing trajectory planning problem can be addressed through trajectory optimization methods based on optimal control theory, which

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essentially involves discretizing the continuous optimal control problem (OCP) and transforming it into a finite-dimensional nonlinear optimization problem (NLP). This approach was first proposed by Shouji et al. (1992). Martinsen et al. (2020) proposed a berthing trajectory planning method based on trajectory optimization, which formulates it as a nonlinear OCP to solve for collision-free trajectories. Maki et al. (2020) introduced an evolution strategy based on the covariance matrix adaptation (CMA-ES), which modeled the OCP as a minimum time cost problem to obtain near-global optimal trajectories. However, CMA-ES cannot be directly applied to efficient planning of berthing trajectories, as it typically requires a significant amount of time to solve. Additionally, these methods have only been validated in obstacle-free waters or ports with simple shapes, neglecting the actual shape of the dock and the distribution of obstacles. Building on the aforementioned literature, Rachman et al. (2022) utilized the offline solutions obtained by CMA-ES as a warm start for semi-online trajectory optimization, considering polygonal docks, and employed the point-in-polygon method to generate necessary spatial constraints to ensure no collisions occur within non-convex areas. However, for more complex-shaped ports, the point-in-polygon method may complicate the spatial constraints of the optimization model, thereby reducing the solution efficiency. Miyauchi et al. (2022) supplemented the functionality of CMA-ES by considering constraints of terrain factors and the influence of constant wind, proposing a ship domain collision avoidance algorithm to calculate collision-free optimal trajectories, but this method further increases the computational burden.

Most current trajectory planning and optimization approaches ignore the rudder effect of the USV's low-speed motion, which may cause the terminal part of the planned berthing trajectory to exceed the USV's dynamic constraints. Moreover, the OCP-based trajectory planning algorithm often consume a long computing time, hence, how to improve the solving efficiency of berthing trajectories still remains to be addressed.

By pre-planning the berthing trajectory, the berthing control problem can be solved in real time by simply tracking the reference trajectory. However, tracking the pre-planned trajectory is not easy. The USVs possess typical characteristics of nonlinearity, large time-delay, high inertia, and under-actuation, which result in poor maneuverability during low-speed navigation, significantly impacting the accurate control of USV berthing (Wang et al., 2022). Moreover, the effect of environmental disturbance increases at low-speed (Wakita et al., 2023). Therefore, developing a tracking controller capable of stably and precisely tracking a given berthing trajectory presents a challenging task.

Significant research has been conducted on the trajectory tracking control of USVs. Among these, model predictive control (MPC) has been extensively studied for its application in berthing control and has demonstrated outstanding performance (Han et al., 2022; Wang et al., 2022; Zhang et al., 2023; Yuan et al., 2023; Lee et al., 2024). However, most studies have overlooked the uncertainty factors in control. MPC relies on the accuracy of the mathematical model of the USV, but in practice, there is a certain deviation between the behavior of the mathematical model and the actual ship. This deviation cannot be ignored in trajectory tracking control, especially in the complex task of automatic berthing. Although MPC can overcome this issue to some extent due to its powerful real-time rolling optimization capability, it is evident that this factor should be considered in simulations.

1.2. Major contributes

Traditional offline OCP approach for berthing trajectory planning needs to solve many nonlinear constraints, resulting in high computational complexity and slow solution speed. Based on the predicted trajectory approach (PTA) proposed in our previous work (Han et al., 2021), a grid-based reverse search strategy (RPTA) for berthing trajectories is introduced to constrain the state of the last waypoint in the berthing trajectory. Considering the impact of environmental loads, the terminal berthing trajectory is reconstructed using the OCP method, with the trajectory planned by the RPTA serving as the initial guess of the optimization. The combination of grid-based RPTA and OCP not only facilitates the modeling of complex port terrains but also ensures that the terminal trajectory fully complies with the USV's dynamic constraints, addressing the maneuverability problem of the USV at low speed. More importantly, the hybrid method overcomes the drawback of slow solution speeds associated with traditional offline OCP method. A nonlinear MPC method is proposed for real-time tracking of the reference trajectory of berthing, ensuring that the USV can accurately dock at the desired berth and achieve the expected heading. Key contributions of this study encompass:

(1) To enhance the solution speed of offline trajectory planning, an OCP-based method for reconstructing berthing trajectories is proposed, taking into account the dynamics of USVs and the impact of environmental loads.
(2) Clarifying the nonlinear relationship between USV speed and rudder effect, constraints are integrated into trajectory planning and tracking algorithms to address the challenge of poor maneuverability at low speeds.
(3) A nonlinear MPC controller is developed to evaluate the adaptability of berthing trajectories formulated by different planning algorithms, while concurrently addressing the control uncertainties inherent to USV berthing operations.

This paper is organized as follows: Section 2 primarily introduces the environmental modeling and mathematical modeling of USVs. In Section 3, RPTA and OCP methods are proposed for berthing trajectory planning and reconstruction. Section 4 designs a nonlinear MPC controller for trajectory tracking. Simulations are presented and discussed in Section 5. Section 6 provides concluding remarks regarding our work.

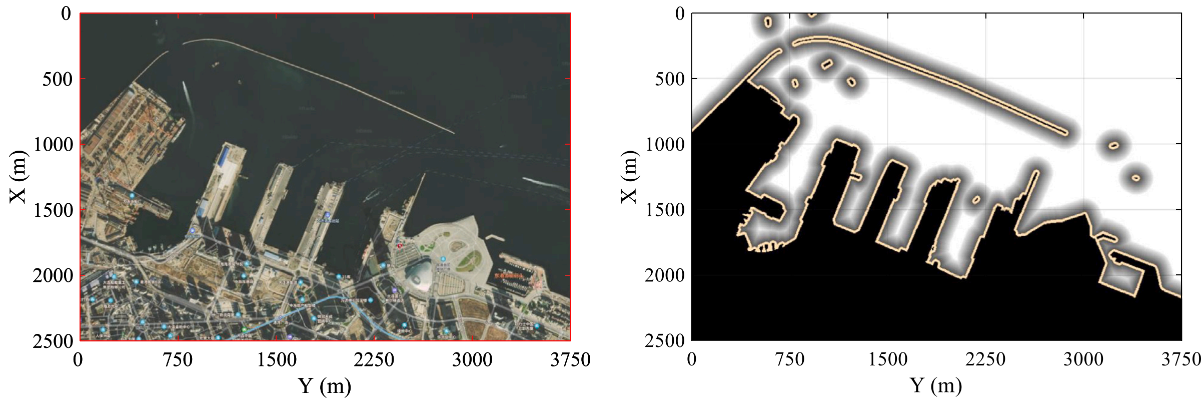
2. Environmental model and USV model

2.1. Environmental model

Nautical navigation typically relies on electronic charts, traditional paper maps, and satellite imagery. This study leverages satellite imagery to construct a detailed global map that includes essential environmental features. Fig. 1(a) presents a satellite imagery of Dalian Port, designated as the simulation testing site for automatic berthing. The area features a complex topography, with the construction of multiple wharves and breakwaters. To further increase the complexity of the site, some ships that may be sailing in the Dalian Port will also be modeled as obstacles.

The Grid Map technique, frequently used in the fields of robotics, segments the satellite image into a grid map, where cells are either unobstructed (white) or obstructed (black). The discretized grid map of Dalian Port is depicted in Fig. 1(b), with a grid resolution of 500×750, which corresponds to a side length of 5 meters for each grid cell. For the purpose of safe USV navigation, the obstructed cells are augmented with additional obstructed cells. These augmented cells, depicted in orange-yellow, are treated as obstructed for global trajectory planning. As the USV approaches the obstructed obstacles, its navigational risks will increase. Therefore, classifying the navigable waters into risk levels based on the distribution of obstacles is of significant importance for the safety of USV. The non-uniform cell-cost is employed to denote the traversability of the grid map. The cost for obstructed cells is set to infinity, while the cost for the remaining unobstructed cells is assessed by the following risk function:

$$R(c) = \begin{cases} 1, & d > D_L \\ \kappa \cdot (D_L - d)^2 + 1, & d \leq D_L \end{cases} \tag{1}$$



(a) satellite imagery (b) grid map

Fig. 1. The satellite map of Dalian Port is discretized into a grid map.

where d denotes the straight-line distance between an unobstructed cell and its nearest obstructed cell. κ represents the weight coefficient. And D_L signifies the boundary range of the risk area, which is determined by the user based on coastal complexity, collision risk assessment, and the maneuverability of the USV. As depicted in Fig. 1(b), the grid map generates a layer of risk cells with decreasing grayscale values near obstacles.

2.2. USV model

The mathematical framework encapsulates the dynamics of the USV's horizontal movement, which encompasses the surge, sway, and yaw behaviors influenced by the propulsion system, rudder mechanism, and environmental loads. The correspondence between the USV's spatial orientation within a geodetic coordinate system and its velocity in the body-fixed coordinate system is designed via a matrix of transformation (Fossen, 2011):

$$\dot{\boldsymbol{\eta}} = \boldsymbol{T}(\psi)\boldsymbol{v} \tag{2}$$

where the vector $\boldsymbol{\eta} = [x, y, \psi]^{\mathrm{T}}$ delineates the USV's displacement in the geodetic frame, encompassing its horizontal positions and yaw orientation. The velocity vector $\boldsymbol{v} = [u, v, r]^{\mathrm{T}}$ captures the linear and rotational velocities of the USV in the body-fixed frame. And $\boldsymbol{T}(\psi)$ can be expressed as:

$$\boldsymbol{T}(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3}$$

To emulate the motion responses subject to the influence of environmental loads and control inputs, the generalized equations of motion endowed with three degrees of freedom are employed:

$$\boldsymbol{M}_{RB}\dot{\boldsymbol{v}} + \boldsymbol{M}_A\dot{\boldsymbol{v}}_r + \boldsymbol{C}_{RB}(\boldsymbol{v})\boldsymbol{v} + \boldsymbol{C}_A(\boldsymbol{v}_r)\boldsymbol{v}_r + \boldsymbol{D}(\boldsymbol{v}_r)\boldsymbol{v}_r = \boldsymbol{\tau} + \boldsymbol{\tau}_{\mathrm{wind}} \tag{4}$$

where the matrix $\boldsymbol{M}_{RB}$ is designated to represent the inertial properties of the rigid body. The term $\boldsymbol{C}_{RB}(\boldsymbol{v})$ encapsulates the effects of Coriolis and centripetal forces. The matrix $\boldsymbol{M}_A$ accounts for the hydrodynamic added mass, while $\boldsymbol{C}_A(\boldsymbol{v}_r)$ corresponds to the matrix of Coriolis and centripetal forces attributed to this added mass. The matrix $\boldsymbol{D}(\boldsymbol{v}_r)$ embodies the damping forces. The term $\boldsymbol{v}_r$ signifies the vector of relative velocities with respect to the environment. The vector $\boldsymbol{\tau} = [\tau_u, \tau_v, \tau_r]^{\mathrm{T}}$ denotes the vector of control forces exerted by the USV's actuators, and $\boldsymbol{\tau}_{\mathrm{wind}}$ represents the forces exerted by wind upon the USV. Considering the underactuation characteristics, the sway force $\boldsymbol{\tau}_v$ of the USV is considered negligible in this paper.

3. Hybrid trajectory planning approach

This section mainly introduces a grid-search and optimal control integration approach for trajectory planning of automatic berthing. This approach initially applies Predicted Trajectory Approach (PTA) in a reverse manner to obtain the preliminary berthing trajectory. Subsequently, it employs an OCP-based method to reconstruct the terminal part of berthing trajectory. This reconstruction ensures that the resulting trajectory fully adheres to the dynamic constraints of the USV, guaranteeing its traceability under the influence of external environmental loads.

3.1. Reverse search by predicted trajectory approach

PTA is an efficient trajectory planning method that was proposed in our previous work (Han et al., 2021). It is a typical grid-search method, fundamentally a variant of the A* algorithm. Unlike the traditional A* algorithm which searches through the grids, PTA searches for waypoints within predicted trajectories. PTA is capable of considering the dynamic constraints of the USV to plan a global trajectory from the starting point to the goal point. Nevertheless, it cannot constrain the state at the end of the berthing trajectory (i.e., the berthing waypoint), and thus, improvements are necessary. The method for improvement is known as Reverse Search by Predicted Trajectory Approach, referred to as RPTA. RPTA assumes that the trajectory planned in the forward direction can be tracked in reverse by the vessel, meaning that the trajectory planned in the forward direction still complies with the vessel's dynamic constraints when tracked in reverse. Once the position of the target berth and the desired berthing heading are determined, the initial state and the position of goal for RPTA are defined as follows:

$$\boldsymbol{x}_0 = [x, y, \psi, u, v, r]^{\mathrm{T}} = [x_g, y_g, \psi_d - \pi, 0, 0, 0]^{\mathrm{T}} \tag{5}$$

$$\boldsymbol{P}_g = [x_0, y_0]^{\mathrm{T}} \tag{6}$$

where $[x_g, y_g]^{\mathrm{T}}$ is the position of the target berth. ψ_d is the desired berthing heading, taking care to note that the initial heading in RPTA is opposite to the desired berthing heading. $[x_0, y_0]^{\mathrm{T}}$ is the position of the USV when it began its berthing mission. By conducting a reverse search from the target berthing position, RPTA ensures that the terminal velocity of the obtained berthing trajectory is zero, with the heading aligned with the desired berthing heading.

Considering the physical characteristics of the rudder mechanism, RPTA initially identifies two sets:

$$\boldsymbol{M}_d = \left\{\tau_r \in \boldsymbol{R} \mid \tau_r \in \left[\tau_{rc} - \Delta\tau_r, \tau_{rc} + \Delta\tau_r\right]\right\} \tag{7}$$

$$\boldsymbol{M}_f = \left\{\tau_r \in \boldsymbol{R} \mid \tau_r \in \left[\tau_{r,\min}, \tau_{r,\max}\right]\right\} \tag{8}$$

where the set $\boldsymbol{M}_d$ represents all yaw moments that the USV can achieve within a unit sampling intervalΔT, and the set $\boldsymbol{M}_f$ denotes a collection constituted by the extreme yaw moments. The τ_{rc} signifies the current yaw moment. The $\Delta\tau_r$ represents the maximum allowable change in yaw moment over the unit sampling interval. $\tau_{r,\min}$ and $\tau_{r,\max}$ correspond to the minimum and maximum yaw moments that the rudder mechanism can provide, respectively.

The search space $\boldsymbol{M}_s$ is defined as the intersection of $\boldsymbol{M}_d$ and $\boldsymbol{M}_f$:

$$\boldsymbol{M}_s = \boldsymbol{M}_d \cap \boldsymbol{M}_f \tag{9}$$

Subsequently, the search space is discretized into a set of feasible yaw moments by the given interval $\Delta\tau_{r,h}$. Each set of feasible yaw moments corresponds to an identical surge force. Considering that the speed of the USV gradually decreases during docking, the surge force is defined to increase slowly in the RPTA search until it reaches the maximum allowed value. The input of current surge force is defined as:

$$\tau_{uc} = \min\left\{\tau_{u,\max}, 0.1 + \frac{n_T}{50}\tau_{u,\max}\right\} \tag{10}$$

where $\tau_{u,\max}$ denotes the maximum value of the surge force, and n_T represents the cumulative number of sampling intervals in the trajectory planning.

By feeding each set of yaw moments and the corresponding identical surge force into the mathematical model of the USV, the predicted trajectories can be obtained. For each predicted trajectory, the corresponding yaw moment and surge force are maintained constant over the prediction period T_p.

All predicted trajectories are decomposed into waypoints based on the unit sampling interval ΔT. If a waypoint on a predicted trajectory is located within an obstructed cell or if two adjacent waypoints are blocked by an obstructed cell, the predicted trajectory where the waypoint resides is considered to pose a collision risk and is discarded. The *Line-of-Sight* function (Daniel et al., 2010) is utilized to ascertain whether two adjacent waypoints on the grid map are obstructed by obstacles. Considering the risk cost of cells near obstacles, the objective function of RPTA is defined as follows:

$$F(c) = G(c) + H(c) + \lambda \cdot \sum R(c) \tag{11}$$

where λ is the weighting coefficient. $G(c)$ represents the actual time cost from the initial state $\boldsymbol{x}_0$ to the waypoint found so far. $H(c)$ represents the heuristic cost from the waypoint to the goal position $\boldsymbol{P}_g$. $\Sigma R(c)$ is the cumulative risk value from the initial state x_0 to the waypoint according to Eq. (1). $F(c)$ is the estimated cost from the initial state $\boldsymbol{x}_0$ via the waypoint to the goal position $\boldsymbol{P}_g$.

By means of an example, a more intuitive understanding of how to use the RPTA to plan a docking trajectory, while satisfying the dynamic constraints of the USV, is provided, as shown in Fig. 2.

(a) Iteration 0: The starting coordinate (i.e., the position of the target berth) is set to L1 (green triangle), and the goal coordinate is A12 (green pentagram). At the starting point, the initial USV state $\boldsymbol{x}_0$ and initial control input $\boldsymbol{\tau}_0$ are known.

(b) Iteration 1: The starting point is selected as the Current WayPoint (CWP, cell it occupies is filled with red) , and the search space is obtained according to Eq. (7)–(9). By discretizing the search space and solving the Eq. (10), yaw moments and their corresponding surge force are respectively derived and input into the mathematical model of the USV, yielding all predicted trajectories. If a predicted trajectory collides with obstacles or exceeds boundaries, the corresponding trajectory is discarded (green dashed trajectory), and the remaining predicted trajectories are decomposed into waypoints. The cost of each waypoint is then calculated based on Eq. (11). If multiple waypoints are located in the same cell (e.g., K2), the waypoint with the smallest cost is selected to match the cell and added to the *Open_List* (the corresponding waypoint turns blue, and the cell turns yellow), thus achieving a one-to-one correspondence between waypoints and cells. Cells also store the sequence of motion state and control input information from the CWP to the waypoint, facilitating backward retrieval of feasible trajectories from the CWP to the starting point. Finally, the CWP is added to the *Closed_List* (the cell storing the CWP will turn blue in the next iteration).

(c) Iteration 2: The waypoint with the smallest cost in the *Open_List* is selected as the new CWP. The search space is obtained through the motion state and control input of the CWP, which leads to the generation of predicted trajectories. Then, the newly generated waypoints are obtained and their cost is calculated. If a newly generated waypoint resides in a cell that had a waypoint added to the *Open_List* in previous iterations, and its cost is smaller than the previous one, the previous waypoint will be replaced by the newly generated one, ensuring a one-to-one match. The motion state and control input information stored in this cell will also be updated. For example, if the newly generated waypoint in cell I6 has a smaller cost than the one saved in Iteration 1, the previously stored waypoint in the *Open_List* for cell I6 will be replaced by the newly generated one.

(d) Iteration 5: It can be observed that two newly generated trajectories have bypassed the obstacles and are gradually approaching the goal point.

(e) Iteration 17: All predicted trajectories collide with obstacles or exceed boundaries. This implies that no new waypoints will be generated in this iteration, and the *Open_List* will not be updated. Subsequently, the RPTA needs to determine whether the *Open_List* is empty. If the *Open_List* is empty, then no feasible berthing trajectory can be found. If not, the waypoint with the smallest cost is continued to be selected from the *Open_List* as the new CWP, and the iteration continues.

(f) Iteration 20: A new waypoint is generated on the goal cell A12, indicating that the RPTA has found a feasible trajectory from the starting point to the goal point. However, this trajectory may not be optimal.

(g) Iteration 29: The newly generated waypoint in the target cell A12 has a smaller cost than before, indicating that the RPTA has found a more efficient trajectory.

(h) Iteration 36: When the CWP is located in the target cell A12, it signifies that the RPTA has found the optimal trajectory, and the iteration ends. Finally, the planned trajectory (red solid line) is constructed backward through the sequence of motion state information saved in the cells.

Ultimately, by employing this backward search method from the docking target point to the docking starting point, a preliminary docking reference trajectory can be obtained, which can be expressed as:

$$\boldsymbol{p}_R = [\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3, \ldots, \boldsymbol{p}_N]^{\mathrm{T}} \tag{12}$$

where N denotes the number of waypoints that constitute the reference trajectory planned by RPTA. $\boldsymbol{p}_1 = [x_0, y_0]^{\mathrm{T}}$ represents the starting point from which the USV commences its berthing task. $\boldsymbol{p}_N = [x_g, y_g]^{\mathrm{T}}$ signifies the position of the target berth. In this context, due to the nature of the reverse search, the docking reference trajectory $\boldsymbol{p}_R$ encompasses only the positional information of the waypoint, without any heading or velocity information.

3.2. OCP-based trajectory reconstruction

Through the RPTA, a preliminary berthing trajectory can be obtained. However, this trajectory fails to account for the influence of environmental loads since it is derived from a reverse search. Consequently, the trajectory obtained by RPTA is unreliable, particularly during the low-speed navigation of vessels.

The OCP method is an optimization approach. The berthing trajectory planning problem can be addressed through the OCP method, which essentially involves discretizing the continuous OCP and transforming it into a finite-dimensional NLP. However, the OCP method necessitates a well-founded initial guess to initiate the optimization process (Kelly, 2017). An inappropriate initial guess not only may fail to

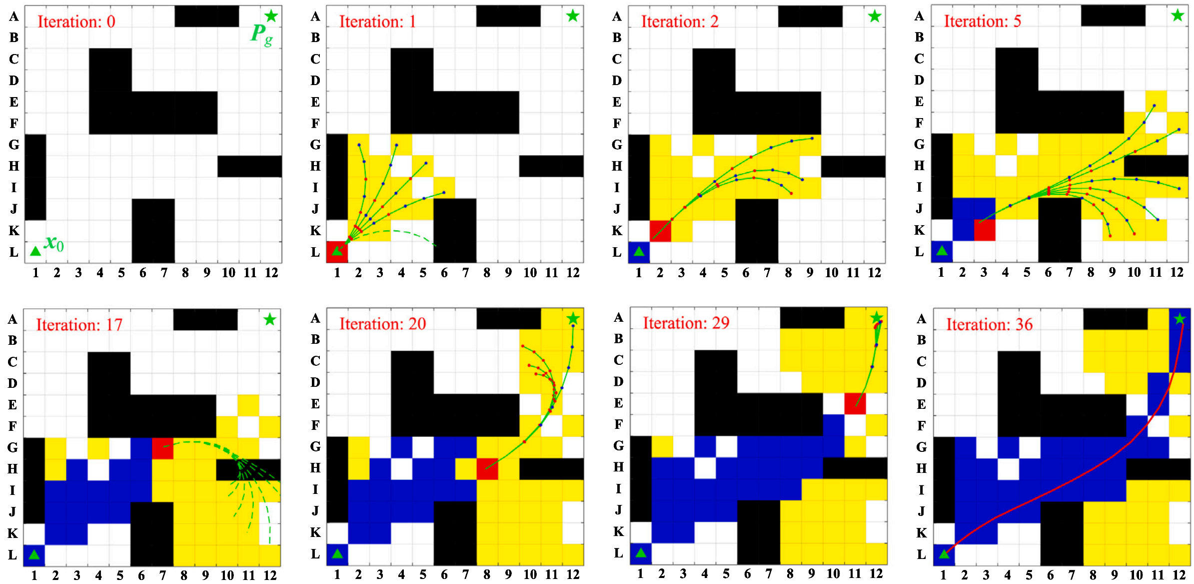


Fig. 2. Example trace of RPTA.

solve the optimization problem, but also may increase the computation time required to solve the berthing trajectory.

In fact, the berthing trajectory during the normal speed phase does not experience a decrease in its trackability due to the influence of environmental factors. This is because the driving force of the vessel at the normal speed phase is sufficient to counteract the impact of environmental loads. This perspective has been validated by the simulation results detailed in Section 5. Considering the computational efficiency issues of the OCP method, only the terminal part of the berthing trajectory obtained by the RPTA is reconstructed. The unreconstructed portions retain the results derived from RPTA. The last M waypoints in RPTA trajectory are taken for reconstruction, and the reconstructed part in Eq. (12) can be expressed as:

$$\boldsymbol{P}_{RE} = \left[\boldsymbol{P}_{N-M+1}, \boldsymbol{P}_{N-M+2}, \dots, \boldsymbol{P}_{N}\right]^{\mathrm{T}} \tag{13}$$

Taking $\boldsymbol{x} = [x, y, \psi, u, v, r]^{\mathrm{T}}$ as the state variable, and $\boldsymbol{u} = [\tau_u, 0, \tau_r]^{\mathrm{T}}$ as the control input, the dynamics of the USV motion in Eq. (4) can be expressed as

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) = \begin{bmatrix} \boldsymbol{T}(\psi)\boldsymbol{v} \\ \left(\boldsymbol{M}_{RB} + \boldsymbol{M}_{A}\right)^{-1}\left(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} - \boldsymbol{C}_{RB}(\boldsymbol{v})\boldsymbol{v} - \boldsymbol{C}_{A}(\boldsymbol{v}_r)\boldsymbol{v}_r - \boldsymbol{D}(\boldsymbol{v}_r)\boldsymbol{v}_r\right) \end{bmatrix} \tag{14}$$

Here, we will employ a common objective function: the integral of control effort squared. This cost function is desirable as it tends to yield smooth solution trajectories that are computationally straightforward to derive. Using the trapezoidal collocation method, the trajectory optimization problem is stated as a nonlinear program: determine

$$\begin{aligned} \boldsymbol{X} \equiv\ & \left[[x_1, \dots, x_M], [y_1, \dots, y_M], [\tau_{u,1}, \dots, \tau_{u,M}],\right. \\ & \left.[\tau_{r,1}, \dots, \tau_{r,M}], \psi_{\mathrm{f}}, u_{\mathrm{f}}, v_{\mathrm{f}}, r_{\mathrm{f}}, t_h\right]^{\mathrm{T}} \in \boldsymbol{R}^{4M+5} \end{aligned} \tag{15}$$

that min.

$$J_P = \sum_{k=1}^{M-1} \frac{t_h}{2}\left(\tau_{u,k}^2 + \tau_{r,k}^2 + \tau_{u,k+1}^2 + \tau_{r,k+1}^2\right) \tag{16}$$

subject to

$$\frac{1}{2} t_h(\boldsymbol{f}_{k+1} + \boldsymbol{f}_k) = \boldsymbol{x}_{k+1} - \boldsymbol{x}_k, \quad k \in 1, \dots, M \tag{17}$$

$$dist\left[P(x_k, y_k), L_{\text{berth}}\right] \le B/2, \quad k \in 1, \dots, M \tag{18}$$

$$\tau_{u,\min} \le \tau_{u,k} \le \tau_{u,\max}, \quad k \in 1, \dots, M \tag{19}$$

$$\tau_{r,\min} \le \tau_{r,k} \le \tau_{r,\max}, \quad k \in 1, \dots, M \tag{20}$$

$$-\Delta\tau_u \le \tau_{u,k+1} - \tau_{u,k} \le \Delta\tau_u, \quad k \in 1, \dots, M-1 \tag{21}$$

$$-\Delta\tau_r \le \tau_{r,k+1} - \tau_{r,k} \le \Delta\tau_r, \quad k \in 1, \dots, M-1 \tag{22}$$

$$\begin{aligned} x_1 &= \boldsymbol{P}_{N-M+1}(1), y_1 = \boldsymbol{P}_{N-M+1}(2), x_M = x_g, y_M = y_g, \\ \psi_{\mathrm{f}} &= \psi_d, u_{\mathrm{f}} = 0, v_{\mathrm{f}} = 0, r_{\mathrm{f}} = 0 \end{aligned} \tag{23}$$

where $\boldsymbol{X}$ represents a finite set of decision variables. $[x_1, \dots, x_M]^{\mathrm{T}}$ and $[y_1, \dots, y_M]^{\mathrm{T}}$ are the x-axis and y-axis position variables of the reconstructed trajectory, respectively; $[\tau_{u,1}, \dots, x_{u,M}]^{\mathrm{T}}$ and $[\tau_{r,1}, \dots, x_{r,M}]^{\mathrm{T}}$ correspond to the surge force and yaw moment variables of different waypoints, respectively; ψ_{f}, u_{f}, v_{f} and r_{f} represent the heading, surge velocity, sway velocity, and yaw velocity variables of the last waypoint in the reconstructed trajectory, respectively; $t_h = t_{k+1} - t_k$ represents the duration of two adjacent waypoints. Eq. (16) is the objective function with the accumulation of control effort squared along the entire reconstructed trajectory. Eq. (17) is the constraint of the system dynamics, which are typically nonlinear and describe how the system changes in time. Eq. (18) imposes a constraint on the distance between the USV position and berthing line, ensuring that collisions do not occur. B represents the width of the vessel. Eq. (19)–(22) belong to the path constraints, which enforces restrictions of control effort and its increment. $\Delta\tau_u$ and $\Delta\tau_r$ signify the maximum allowable variation in the surge force and yaw moment per duration t_h, respectively. Eq. (23) denotes the boundary constraint, which puts restrictions on the initial and final states of the system.

During actual navigation, the effect of the USV's rudder diminishes, or even disappears, as the speed decreases. However, few scholars have considered imposing constraints on rudder effect, which is extremely important in berthing trajectory planning and control. The lower and upper bounds of the yaw moment in this paper are dynamically adjusted according to the changes in USV velocity, as illustrated in the following equation:

$$\begin{aligned} \tau_{r,\min} &= \max\left\{\hat{\tau}_{r,\min}, \alpha \cdot V^2 \cdot \hat{\tau}_{r,\min}\right\} \\ \tau_{r,\max} &= \min\left\{\hat{\tau}_{r,\max}, \alpha \cdot V^2 \cdot \hat{\tau}_{r,\max}\right\} \end{aligned} \tag{24}$$

where α is a weighting constant. $\hat{\tau}_{r,\min}$ and $\hat{\tau}_{r,\max}$ are the lower and upper values of the yaw moment at the normal speed, respectively. V is the USV speed.

The nonlinear programming solver requires an initial guess. Taking Eq. (13) as an important component of solving nonlinear programming problems, the initial guess is defined as:

$$\boldsymbol{X}_0 = \left[\boldsymbol{p}_{N-M+1}(1), \ldots, \boldsymbol{p}_N(1), \boldsymbol{p}_{N-M+1}(2), \ldots, \boldsymbol{p}_N(2), \boldsymbol{0}_{1\times M}, \boldsymbol{0}_{1\times M}, \psi_d, 0, 0, 0, \Delta T\right]^{\mathrm{T}} \tag{25}$$

Finally, direct transcription solves a trajectory optimization problem by converting it to a nonlinear program. The resulting NLP is solved using an appropriate Matlab FMINCON solver with SQP (sequential quadratic programming). Combining the remaining trajectory at the normal speed in RPTA with the reconstructed trajectory derived from the OCP method, the whole reference trajectory for automatic berthing is given by:

$$\boldsymbol{p}_R = \left[[\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_{N-M}], [\boldsymbol{p}_{R1}, \boldsymbol{p}_{R2}, \ldots, \boldsymbol{p}_{RM}]\right]^{\mathrm{T}} \tag{26}$$

where $[\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_{N-M}]^{\mathrm{T}}$ is the remaining trajectory at the normal speed in RPTA, while $[\mathbf{p}_{R1}, \boldsymbol{p}_{R2}, \ldots, \boldsymbol{p}_{RM}]^{\mathrm{T}}$ is the reconstructed trajectory derived from the OCP method.

4. Nonlinear MPC for trajectory tracking

In this section, a nonlinear MPC controller is formulated for trajectory tracking of automatic berthing. After the Eq. (14) is discretized, the state space model can be expressed as:

$$\boldsymbol{x}(k+1) = \boldsymbol{f}\left(\boldsymbol{x}(k), \boldsymbol{u}(k)\right) \tag{27}$$

The control inputs of the USV are limited by Eqs. (19)–(22) and (24). Subsequently, the new system state variable for nonlinear MPC is redefined as:

$$\zeta(k) = \begin{bmatrix} \boldsymbol{x}(k) \\ \boldsymbol{u}(k-1) \end{bmatrix} \tag{28}$$

Combining Eqs. (27) and (28), a redefined state-space model is derived as:

$$\begin{aligned} \zeta(k+1) &= \begin{bmatrix} \boldsymbol{f}\left(\boldsymbol{x}(k), \boldsymbol{u}(k-1) + \Delta\boldsymbol{u}(k)\right) \\ \boldsymbol{u}(k-1) + \Delta\boldsymbol{u}(k) \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{f}\left(\boldsymbol{P}_1\boldsymbol{\zeta}(k), \boldsymbol{P}_2\boldsymbol{\zeta}(k) + \Delta\boldsymbol{u}(k)\right) \\ \boldsymbol{P}_2\boldsymbol{\zeta}(k) + \Delta\boldsymbol{u}(k) \end{bmatrix} = \boldsymbol{f}_d\left(\boldsymbol{\zeta}(k), \Delta\boldsymbol{u}(k)\right) \end{aligned} \tag{29}$$

where $\boldsymbol{P}_1 = [\boldsymbol{I}_n, \boldsymbol{0}_{n\times m}]$, $\boldsymbol{P}_2 = [\boldsymbol{0}_{m\times n}, \boldsymbol{I}_m]$. m is the dimension of control input, and n is the dimension of system state. For USV dynamic model, $m = 3$ and $n = 6$.

To achieve precise and consistent tracking of the berthing trajectory by the USV, it is crucial to establish an objective function that accounts for tracking deviation. Therefore, the controllers for trajectory tracking in the USV are effectively delineated by the following optimization problem: that min

$$J_C = \sum_{i=1}^{N_p} \left\|\boldsymbol{P}_3\boldsymbol{x}(k+i) - \boldsymbol{p}_R(k+i)\right\|_Q^2 + \left\|\boldsymbol{P}_4\boldsymbol{x}(k+N_p) - \psi(k+N_p)\right\|^2 \tag{30}$$

subject to

$$\begin{aligned} &\boldsymbol{\zeta}(k+i+1) = \boldsymbol{f}_d\left(\boldsymbol{\zeta}(k+i), \Delta\boldsymbol{u}(k+i)\right), i = 0, 1, 2, \ldots, N_p - 1 \\ &\boldsymbol{x}(k+i) = \boldsymbol{P}_1\boldsymbol{\zeta}(k+i), i = 1, 2, 3, \ldots, N_p \\ &\Delta\boldsymbol{U}_{\min} \le \Delta\boldsymbol{U}(k) \le \Delta\boldsymbol{U}_{\max} \\ &\boldsymbol{U}_{\min} \le \boldsymbol{M}\Delta\boldsymbol{U}(k) + \Delta\boldsymbol{U}(k) \le \boldsymbol{U}_{\max} \end{aligned} \tag{31}$$

where index $k+i$ denotes the predicted value at i steps ahead of the current time k. N_p is the length of prediction horizon. The objective function includes the tracking deviation of positions and heading with weighting matrices $\boldsymbol{Q}$. $\boldsymbol{U}_{\min}$ and $\boldsymbol{U}_{\max}$ represent the set of minimum and maximum values of control input in the control horizon, respectively. $\Delta\boldsymbol{U}_{\min}$ and $\Delta\boldsymbol{U}_{\max}$ represent the set of minimum and maximum values of input increments in the control horizon, respectively. $\boldsymbol{M} = \boldsymbol{S}_{N_p} \otimes \boldsymbol{I}_m$, where $\boldsymbol{S}$ is the N_p-th order lower triangular matrix. $\psi(k+N_p)$ is the direction from waypoint $\boldsymbol{P}_R(k+N_p-1)$ to waypoint $\boldsymbol{P}_R(k+N_p)$ in the reference trajectory. $\Delta\boldsymbol{U}(k)$ is expressed as:

$$\Delta U(k) = \begin{bmatrix} \Delta\boldsymbol{u}(k) \\ \Delta\boldsymbol{u}(k+1) \\ \vdots \\ \Delta\boldsymbol{u}(k+N_p-1) \end{bmatrix} \tag{32}$$

$\boldsymbol{P}_3$ and $\boldsymbol{P}_4$ is defined as:

$$\boldsymbol{P}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{P}_4 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \tag{33}$$

In this paper, $N_p = 10$, $Q = \mathrm{diag}[0.1, 0.1]$, $\tau_{u,\min} = -2$, $\tau_{u,\max} = 2$, $\hat{\tau}_{r,\min} = -1.5$, $\hat{\tau}_{r,\max} = 1.5$, $\Delta\tau_u = 0.2$, $\Delta\tau_r = 0.25$. The whole algorithm framework for USV trajectory planning, reconstruction and tracking is shown in Fig. 3.

5. Results and discussion

Numerical simulations are performed to evaluate the proposed trajectory planning, reconstruction and tracking approaches. A 1:70 scale model of a North Sea supply ship, the CyberShip II, is utilized as the representative USV. Accordingly, a consistent scaled grid map of Dalian Port serves as the environment for the experiments. The CyberShip II has a mass of 23.8 kg, length of 1.255 m, and breadth of 0.29 m, with detailed hydrodynamic parameters available in the Ref. Skjetne et al. (2004). All algorithms are implemented in MATLAB 2022b running on an Intel Core i9 CPU at 5.8 GHz with 128 GB RAM.

5.1. Automatic berthing at different berths regardless of environmental loads

Three berthing scenarios are set from Dalian Port, which shares the same scaling ratio as the CyberShip II model. The starting point of Scenario-1 is $P_{s1} = [21, 46]^{\mathrm{T}}$, and the desired berthing spot is $\eta_{d1} = [23.5, 27.6, -24°]^{\mathrm{T}}$. The starting point of Scenario-2 is $P_{s2} = [15, 50]^{\mathrm{T}}$, and the desired berthing spot is $\eta_{d2} = [24, 24, -24°]^{\mathrm{T}}$. The starting point of Scenario-3 is $P_{s3} = [5, 40]^{\mathrm{T}}$, and the desired berthing spot is $\eta_{d3} = [21, 17, 246°]^{\mathrm{T}}$. RPTA+MPC algorithm is used for trajectory planning and tracking control of automatic berthing, and the results are shown in Fig. 4. To present the docking performance of the proposed algorithms in a clear and intuitive manner, the time-consuming of RPTA (T_p), the average iteration time of the MPC (T_c), the berthing position deviation ($P_{e,\mathrm{end}}$), the berthing heading deviation (ψ_e), and the berthing velocity residual (V_e) in various berthing scenarios are all recorded in Table 1.

Table 1
The docking performance in various berthing scenarios.

Condition	T_p (s)	T_c (s)	$P_{e,\mathrm{end}}$ (m)	ψ_e (°)	V_e (m/s)
Scenario-1	3.853	0.209	0.015	5.76	0.0004
Scenario-2	6.911	0.180	0.042	2.04	0.0068
Scenario-3	18.308	0.199	0.033	1.42	0.0047

From Fig. 4 and Table 1, it can be observed that the greater the number of grids in the *Open_List* and *Closed_List*, the longer the time required for the RPTA to search for the global trajectory. Considering that global trajectory planning is performed offline, this computational time is entirely acceptable and, subjectively, commendable. The average iterative time of the MPC refers to the duration necessary for the controller to solve for the optimal control sequence at each iteration step. Compared to the sampling interval ($\Delta T = 0.5$ s) adopted in this study, the MPC is fully capable of achieving real-time rolling optimization

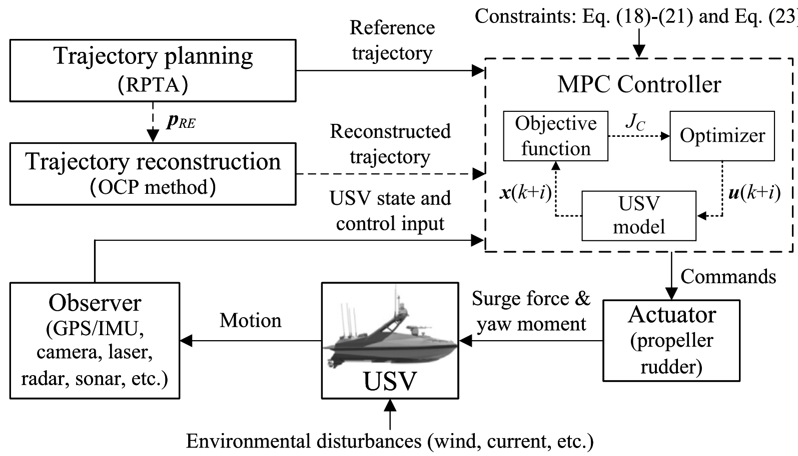


Fig. 3. The whole algorithm framework for USV trajectory planning, reconstruction and tracking.

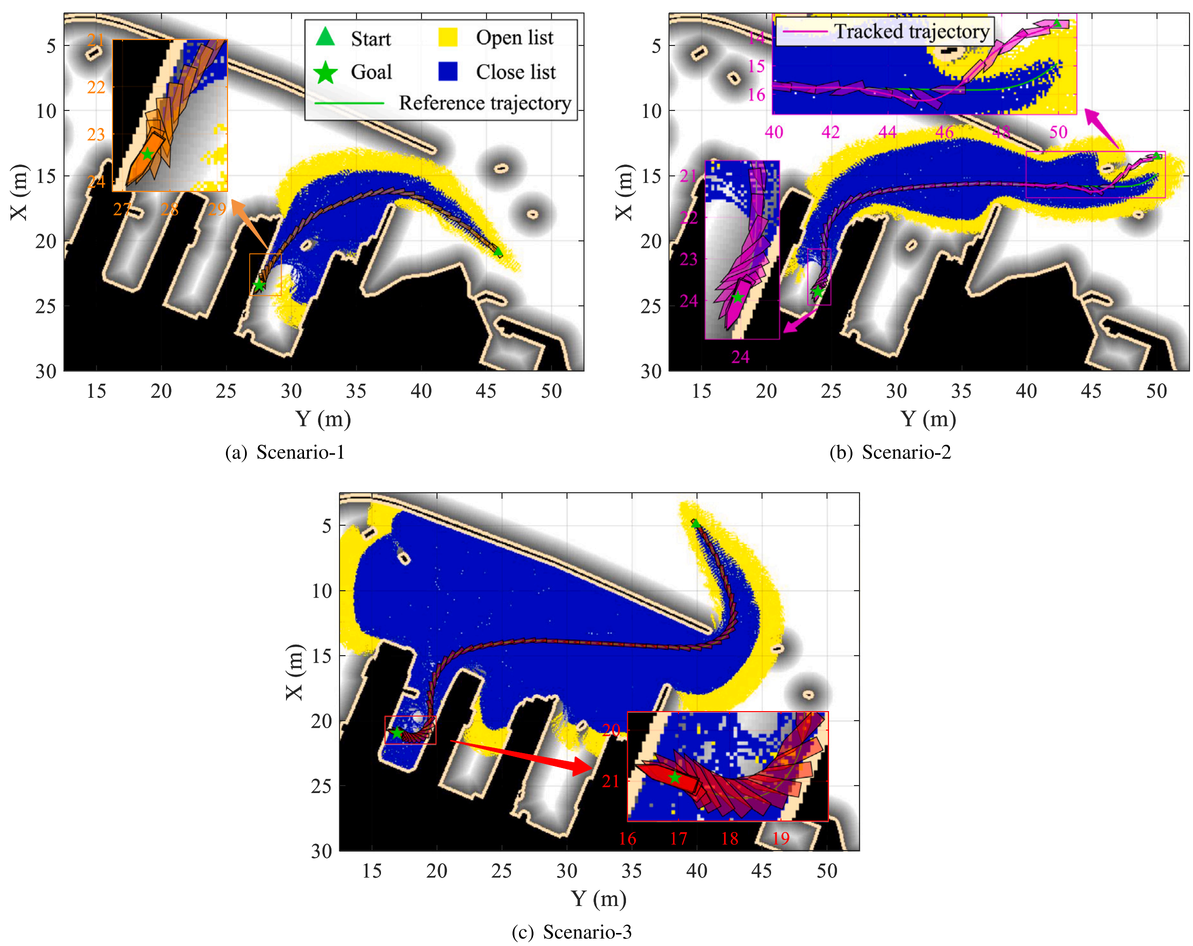


Fig. 4. The results of trajectory planning and tracking control of automatic berthing at different berths.

calculations. Based on the data of berthing position deviation, heading deviation, and velocity residual, the combined strategy of RPTA and MPC demonstrates preferable performance regardless of environmental loads. As can be seen from Fig. 4(b), the initial position of the USV and the starting point of the reference trajectory are set with a significant deviation. This is done with the purpose of testing the robustness of the MPC-based trajectory tracking controller. The results indicate that the controller enables the USV to gradually approach the reference trajectory and eliminate the setting deviation, thereby achieving precise trajectory tracking.

In the partial magnification of all scenarios, a "fluctuation" phenomenon exists in the heading of the USV control at the end of the berthing process. This occurs because the USV increases the angle between the heading and the velocity direction (i.e., increases the velocity component in the sway direction) to enhance the hydrodynamic resistance, thereby achieving rapid deceleration. Consequently, the navigation trajectory that starts from zero and accelerates gradually, if used for reverse tracking, may exceed the dynamic constraints of the USV. Considering the inertia of a ship is typically substantial, its ability to accelerate is superior to its deceleration performance. Therefore, it

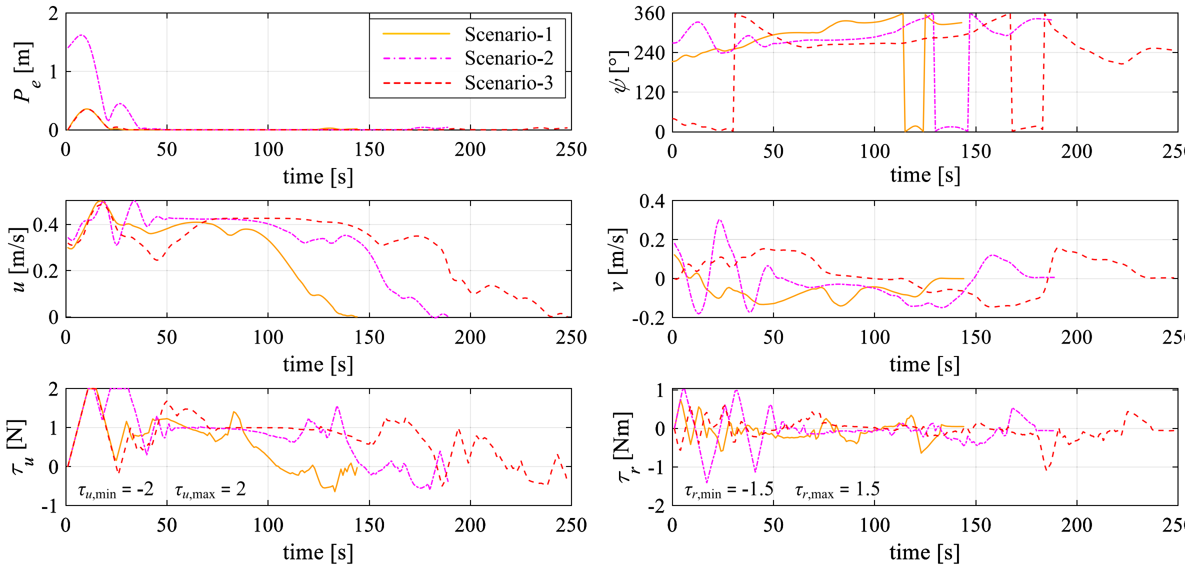


Fig. 5. The position deviation, motion states and control inputs of the USV at different berths.

is imperative to reconstruct the terminal berthing trajectory, especially when environmental loads are taken into account, to enhance the safety of the berthing process. The automatic berthing of the USV considering environmental loads will be conducted and discussed in the subsequent subsection.

Additionally, the tracking position deviation P_e, motion states (including yaw angle ψ, surge velocity u and sway velocity v) and control inputs (including surge force τ_u and yaw moment τ_r) of the USV with various scenarios are supplemented, as depicted in Fig. 5.

In Fig. 5, there is a significant deviation in the initial phase of trajectory tracking, which can be attributed to the inconsistency between the preset initial state of the USV and the reference trajectory. In the figure of the yaw moment over time, the magnitude of the yaw moment at the terminal phase of the berthing process is essentially zero, indicating the rudder effect constraint added in Eq. (24).

5.2. Automatic berthing at different berths considering environmental loads

Three berthing scenarios defined in the previous subsection continue to be used considering the effect of environmental loads. The wind speed is set to V_w = 1.5 m/s, with a wind direction of ψ_w = 45°. RPTA+MPC and OCP+MPC algorithms are executed for automatic berthing, and the results are shown in Fig. 6 and Fig. 7 respectively. In the figure, the legend's reference to "OCP" actually denotes "RPTA+OCP", which signifies the process of acquiring an initial berthing trajectory via RPTA, followed by the reconstruction of its terminal part using the OCP method to derive a novel berthing trajectory.

From Fig. 6, it can be observed that when environmental disturbances occur, using RPTA as the reference trajectory still achieves good tracking performance during the normal speed phase. Compared with Fig. 4, there is no significant change in the tracking deviation during the normal speed phase. The only difference is that the USV's heading tends to face into the wind to resist the its disturbance. However, during the terminal deceleration phase, the effect of the wind on the hull becomes more pronounced as the USV's speed decreases. This leads to a deterioration in the control effect of the USV at the terminal of berthing, making it difficult to complete the automatic berthing task. Compared with the terminal deceleration phase in Fig. 4(a)–(b), the heading of the USV in Fig. 6(a)–(b) is nearly perpendicular to the direction of velocity to counteract the sway force component of the wind acting on the hull.

Fig. 7 employs the OCP method to reconstruct the berthing trajectory, taking into account the USV's dynamic constraints and the influence of environmental loads to ensure the trackability of the reference trajectory. To more intuitively demonstrate the berthing performance of the two methods considering environmental loads, the time it takes for OCP to reconstruct the trajectory ($T_{p,re}$), the average iteration time of the MPC, the berthing position deviation, the berthing heading deviation, and the berthing velocity residual are presented, as shown in Table 2.

Table 2 reveals that the position deviation, heading deviation, and velocity residual for the RPTA+MPC method are all significantly large in the presence of environmental loads. It has not achieved the berthing requirement of "precisely docking at the desired berth and meeting the expected berthing heading". In contrast, the OCP+MPC method exhibits heading deviation, position deviation, and velocity residual that are within reasonable limits, ensuring the reliability of berthing under environmental disturbances. Additionally, compared to Scenario-1 and Scenario-2, the USV is more easily maneuverable in Scenario-3, as it relies more on the control of the surge velocity. Scenario-1 and Scenario-2, however, require control of the sway velocity, which is evidently more challenging for underactuated vessels. Therefore, the final berthing accuracy at Scenario-3 is commonly higher than at Scenario-1 and Scenario-2.

5.3. Automatic berthing considering control uncertainties

To further compare the differences between the RPTA+MPC and OCP+MPC strategies, Scenario-1 is selected for testing. Six different environmental loads are set up: Load 1, V_w = 1.5 m/s, ψ_w = 45°, $\boldsymbol{v}_c$ = $[0,0,0]^{\mathrm{T}}$ m/s; Load 2, V_w = 1 m/s, ψ_w = 180°, $\boldsymbol{v}_c$ = $[0,-0.05,0]^{\mathrm{T}}$ m/s; Load 3, V_w = 1 m/s, ψ_w = −35°, $\boldsymbol{v}_c$ = $[0.05,0,0]^{\mathrm{T}}$ m/s; Load 4, V_w = 1 m/s, ψ_w = −0°, $\boldsymbol{v}_c$ = $[-0.035,0.035,0]^{\mathrm{T}}$ m/s; Load 5, V_w = 2 m/s, ψ_w = 180°, $\boldsymbol{v}_c$ = $[0.1,0,0]^{\mathrm{T}}$ m/s; Load 6, V_w = 2 m/s, ψ_w = −0°, $\boldsymbol{v}_c$ = $[-0.07,0.07,0]^{\mathrm{T}}$ m/s, where the symbols V_w, ψ_w, $\boldsymbol{v}_c$ correspond to the wind speed, wind direction, and the generalized ocean current velocity vector, respectively. Considering that Dalian Port is equipped with a breakwater, the influence of waves is not included here. Furthermore, to verify the robustness of the MPC, we also considered factors of control uncertainty in the algorithm. The uncertainty in control is manifested in this paper as the addition of random noise within the range of 20% to the optimal control output of the MPC (including surge force and yaw moment), as in Eq. (34). This range of variability

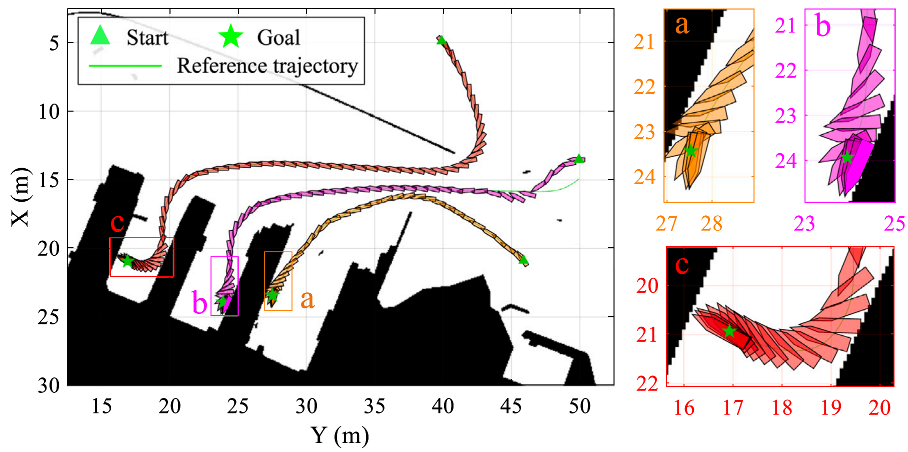


Fig. 6. The results of trajectory planning and tracking control using RPTA+MPC algorithms.

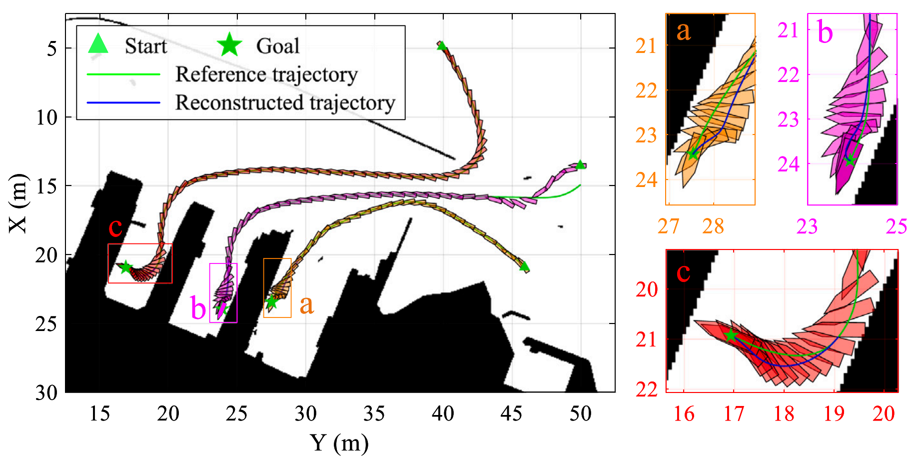


Fig. 7. The results of trajectory planning, reconstruction and tracking control using OCP+MPC algorithms.

Table 2
The docking performance of two algorithms considering environmental loads.

Algorithm	Condition	$T_{p,re}$ (s)	T_c (s)	$P_{e,\mathrm{end}}$ (m)	ψ_e (°)	V_e (m/s)
RPTA+MPC	Scenario-1	–	0.193	0.265	2.28	0.0611
	Scenario-2	–	0.173	0.308	0.06	0.0642
	Scenario-3	–	0.214	0.092	0.02	0.0234
OCP+MPC	Scenario-1	9.889	0.183	0.012	6.65	0.0050
	Scenario-2	60.197	0.181	0.014	1.68	0.0468
	Scenario-3	8.068	0.223	5×10^{-6}	1×10^{-6}	2×10^{-6}

is selected to represent the realistic spectrum of control uncertainties that USVs may encounter under operational conditions. The berthing performance under different environmental loads for RPTA+MPC and OCP+MPC is shown in Table 3.

$$\boldsymbol{u}_w(k) = (1 + \gamma \cdot \omega(k))\boldsymbol{u}(k) \tag{34}$$

where $\boldsymbol{u}_w(k)$ is the control input considering the uncertainty, $\gamma = 20\%$ is the floating limitation of the fluctuations, $\omega(k)$ is a random number in the range of $[-1, 1]$, $\boldsymbol{u}(k)$ represents the control input obtained by the MPC controller.

According to Table 3, the OCP method of reconstructing the trajectory has a significant advantage compared to RPTA, where the average berthing position deviation is reduced by 90.63%, the heading deviation is reduced by 62.05%, and the velocity residual is reduced by 84.38%. This further illustrates that it is necessary to reconstruct the terminal berthing trajectory in the presence of environmental loads, which can ensure the traceability of the reference trajectory and also improve the calculation time of the trajectory planning within an acceptable range.

The Load 1 in Table 3 is consistent with the environmental load set in the previous subsection 5.2. By comparing the data of OCP+MPC algorithm, considering the control uncertainty only increases the trajectory tracking deviation from 0.0388 m to 0.0389 m, which is completely negligible. Moreover, it is even better than the algorithm that ignores control uncertainty in terms of the final berthing position deviation and heading deviation, while the velocity residual is not significantly different between the two. Therefore, even with the influence of uncertain factors, MPC algorithm, relying on its strong robustness

Table 3
The docking performance of two algorithms considering environmental loads and control uncertainties.

Environmental loads	RPTA+MPC			OCP+MPC		
	$P_{e,\mathrm{end}}$ (m)	ψ_e (°)	V_e (m/s)	$P_{e,\mathrm{end}}$ (m)	ψ_e (°)	V_e (m/s)
Load 1	0.255	2.48	0.0596	0.007	3.38	0.0073
Load 2	0.152	0.06	0.0385	9×10^{-4}	0.50	0.0026
Load 3	0.089	7.09	0.0237	8×10^{-4}	0.01	0.0017
Load 4	0.086	0.80	0.0272	3×10^{-4}	0.05	0.0016
Load 5	0.261	16.05	0.1067	0.084	5.63	0.0344
Load 6	0.203	0.45	0.0753	0.005	0.65	0.0041

and real-time rolling optimization capability, can overcome the control deviations caused by uncertain factors and demonstrate excellent performance.

6. Conclusion

The proposed RPTA clearly has advantages in terms of environmental modeling and solution speed. If there is no influence from environmental loads, the trajectory planned by RPTA can be directly used as a reference trajectory to guide the USV to complete automatic berthing and the results such as berthing position deviation, heading deviation and velocity residual fully meet the requirements. This indicates that the forward-planned trajectory can be precisely tracked in reverse by the USV in the absence of environmental load effects. However, the berthing accuracy of RPTA will significantly decrease when there are environmental loads present. Although the trajectory planned by RPTA during the normal speed phase can still be accurately tracked considering environmental loads, the trajectory during the low-speed phase is clearly beyond the USV's dynamic constraints, and the tracking deviation is difficult to eliminate. This is because the maneuvering performance of the USV is poor at low speeds, and the USV cannot provide sufficient driving force to counteract the effects of environmental loads. Therefore, it is necessary to reconstruct the terminal part of the RPTA trajectory when there are environmental loads present.

Employing the OCP method enables the consideration of environmental loads to identify an optimal berthing trajectory that adheres to the USV's dynamic constraints. Trajectory optimization methods based on OCP typically require a good initial guess, and the terminal part of the RPTA trajectory provides just that opportunity. From the simulation results, the reconstructed trajectory can ensure berthing accuracy under the action of environmental loads and control uncertainties. The grid-search and optimal control integration approach retains the high modeling accuracy and fast solution speed while ensuring that the designed berthing trajectory complies with the dynamic constraints of the USV. The combination of RPTA and OCP methods achieves a balance between planning precision and solution efficiency, realizing a complementary advantage, and is feasible for practical application.

The trajectory planning methods proposed in this paper belong to an offline strategy that is only applicable to environments with known static obstacles or dynamic obstacles that can be predicted. However, it is ineffective against unknown static and dynamic obstacles. In our future research, we will focus on dynamic collision avoidance in complex port areas, with a particular emphasis on unknown dynamic obstacles.

CRediT authorship contribution statement

Sen Han: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Lingxiao Yan:** Validation, Methodology, Conceptualization. **Jiahao Sun:** Validation, Methodology. **Shifeng Ding:** Supervision, Funding acquisition. **Fang Li:** Supervision, Funding acquisition. **Feng Diao:** Supervision, Writing – review & editing. **Li Zhou:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Claussmann, L., Revilloud, M., Gruyer, D., Glaser, S., 2019. A review of motion planning for highway autonomous driving. IEEE Trans. Intell. Transp. Syst. 21, 1826–1848.
Daniel, K., Nash, A., Koenig, S., Felner, A., 2010. Theta*: Any-angle path planning on grids. J. Artificial Intelligence Res. 39, 533–579.
Fossen, T.I., 2011. Handbook of Marine Craft Hydrodynamics and Motion Control. John Wiley & Sons.
Han, S., Wang, L., Wang, Y., 2022. A potential field-based trajectory planning and tracking approach for automatic berthing and COLREGs-compliant collision avoidance. Ocean Eng. 266, 112877.
Han, S., Wang, L., Wang, Y., He, H., 2021. An efficient motion planning based on grid map: Predicted trajectory approach with global path guiding. Ocean Eng. 238, 109696.
Higo, Y., Sakano, M., Nobe, H., Hashimoto, H., 2023. Development of trajectory-tracking maneuvering system for automatic berthing/unberthing based on double deep Q-network and experimental validation with an actual large ferry. Ocean Eng. 287, 115750.
Kelly, M., 2017. An introduction to trajectory optimization: How to do your own direct collocation. SIAM Rev. 59, 849–904.
Lee, C., Van Tran, Q., Kim, J., 2024. Safety-guaranteed ship berthing using cascade tube-based model predictive control. IEEE Trans. Control Syst. Technol. 1–8.
Liu, Y., Im, N.-k., Zhang, Q., Zhu, G., 2022. Adaptive auto-berthing control of underactuated vessel based on barrier Lyapunov function. J. Marine Sci. Eng. 10, 279.
Liu, Q., Li, T., Shan, Q., Yu, R., Gao, X., 2021. Virtual guide automatic berthing control of marine ships based on heuristic dynamic programming iteration method. Neurocomputing 437, 289–299.
Liu, H., Peng, Z., Gu, N., Wang, H., Liu, L., Wang, D., 2024. Collision-free automatic berthing of maritime autonomous surface ships via safety-certified active disturbance rejection control. ISA Trans. 148, 24–31.
Maki, A., Sakamoto, N., Akimoto, Y., Nishikawa, H., Umeda, N., 2020. Application of optimal control theory based on the evolution strategy (CMA-ES) to automatic berthing. J. Marine Sci. Technol. 25, 221–233.
Martinsen, A.B., Bitar, G., Lekkas, A.M., Gros, S., 2020. Optimization-based automatic docking and berthing of ASVs using exteroceptive sensors: Theory and experiments. IEEE Access 8, 204974–204986.
Miyauchi, Y., Sawada, R., Akimoto, Y., Umeda, N., Maki, A., 2022. Optimization on planning of trajectory and control of autonomous berthing and unberthing for the realistic port geometry. Ocean Eng. 245, 110390.

Peng, Z., Wang, C., Yin, Y., Wang, J., 2023. Safety-certified constrained control of maritime autonomous surface ships for automatic berthing. IEEE Trans. Veh. Technol. 72, 8541–8552.

Rachman, D.M., Maki, A., Miyauchi, Y., Umeda, N., 2022. Warm-started semionline trajectory planner for ship's automatic docking (berthing). Ocean Eng. 252, 111127.

Shimizu, S., Nishihara, K., Miyauchi, Y., Wakita, K., Suyama, R., Maki, A., Shirakawa, S., 2022. Automatic berthing using supervised learning and reinforcement learning. Ocean Eng. 265, 112553.

Shouji, K., Ohtsu, K., Mizoguchi, S., 1992. An automatic berthing study by optimal control techniques. IFAC Proc. Vol. 25, 185–194.

Skjetne, R., Smogeli, Ø.N., Fossen, T.I., 2004. A nonlinear ship manoeuvering model: Identification and adaptive control with experiments for a model ship. Model. Identif. Control 25, 3–27.

Sun, T., Yin, Y., Liu, C., 2024. Integrated trajectory planning into automatic berthing control of underactuated ship based on fuzzy-backstepping method. Ocean Eng. 291, 116336.

Suyama, R., Miyauchi, Y., Maki, A., 2022. Ship trajectory planning method for reproducing human operation at ports. Ocean Eng. 266, 112763.

Wakita, K., Akimoto, Y., Rachman, D.M., Miyauchi, Y., Maki, A., 2023. Collision probability reduction method for tracking control in automatic docking/berthing using reinforcement learning. J. Marine Sci. Technol. 28, 844–861.

Wang, S., Sun, Z., Yuan, Q., Sun, Z., Wu, Z., Hsieh, T.-H., 2022. Autonomous piloting and berthing based on long short time memory neural networks and nonlinear model predictive control algorithm. Ocean Eng. 264, 112269.

Yuan, S., Liu, Z., Sun, Y., Wang, Z., Zheng, L., 2023. An event-triggered trajectory planning and tracking scheme for automatic berthing of unmanned surface vessel. Ocean Eng. 273, 113964.

Zhang, M., Yu, S.-R., Chung, K.S., Chen, M.-L., Yuan, Z.-M., 2023. Time-optimal path planning and tracking based on nonlinear model predictive control and its application on automatic berthing. Ocean Eng. 286, 115228.